



Thermoelectric properties of ordered monoclinic NaSbSe_2

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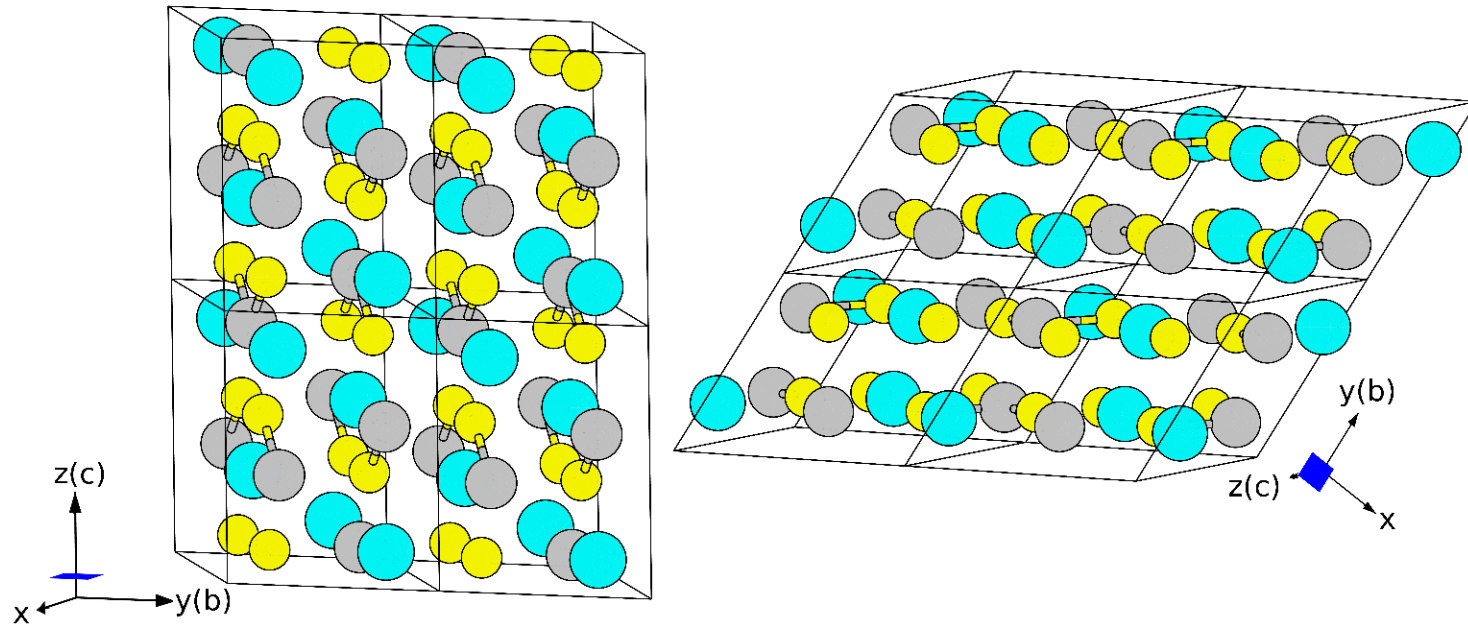
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Introduction

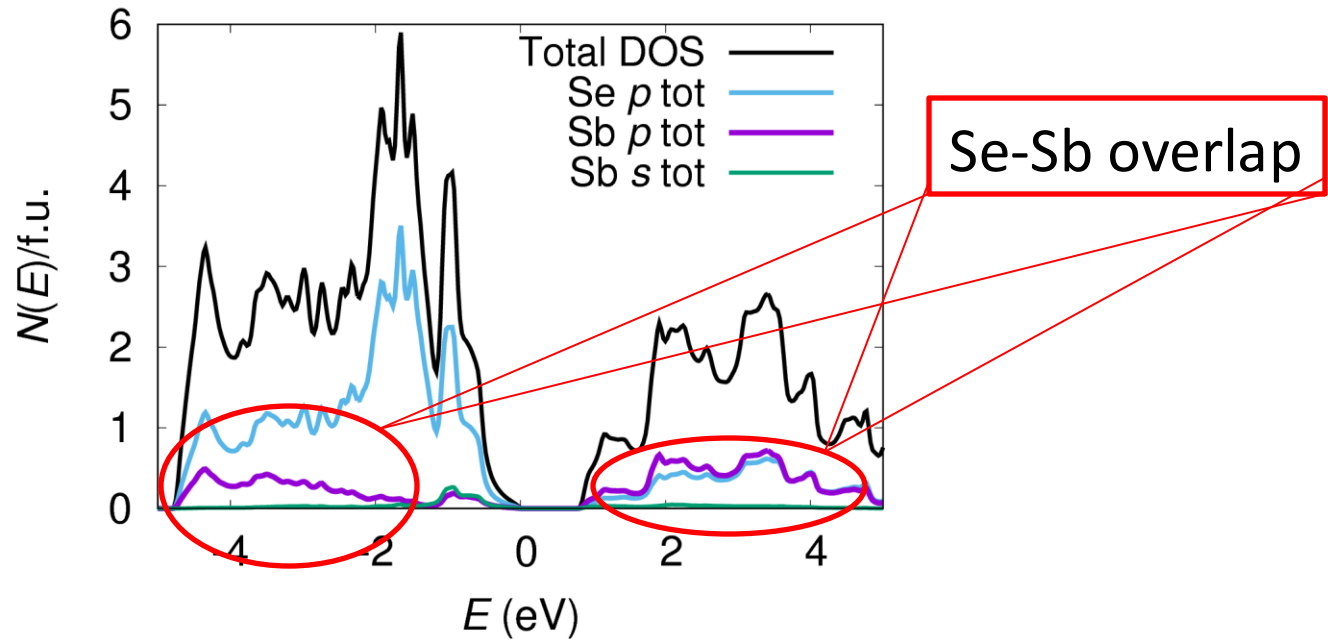
- NaSbSe_2 - *Previously known*
 - Rocksalt structure
 - Cation disordered
 - semi-metallic, low band gap
- NaSbSe_2 - *Recently synthesized*
 - Ordered, monoclinic (C2/c), layered structure
 - Estimated band gap (1.48 eV) for solar absorber
 - Expect good transport behavior

Crystal structure



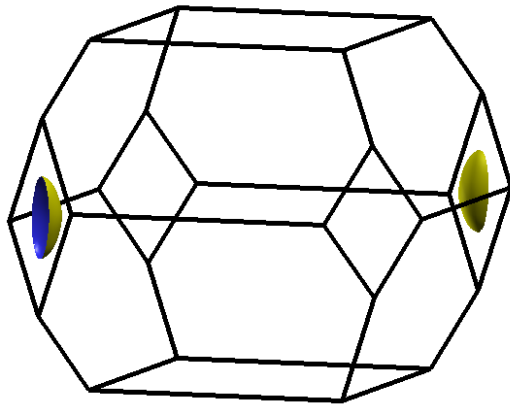
- Monoclinic cell (C2/c, #15)
- Layered structure (NaSbSe₂)
- Se-Sb-Se units, with Na⁺ on either side of Sb³⁺

Electronic structure

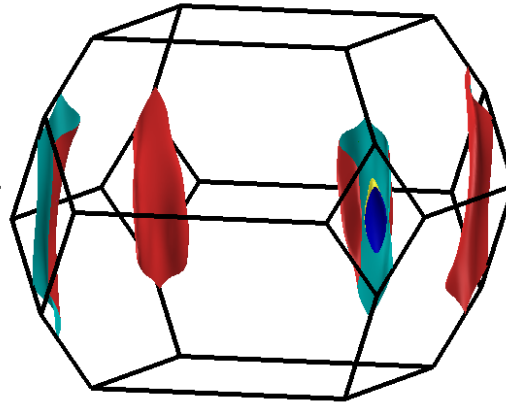


- Obtained gap: 0.84 eV (modified Becke-Johnson)
- Pronounced cross-gap hybridization
Possibility of high Born effective charges

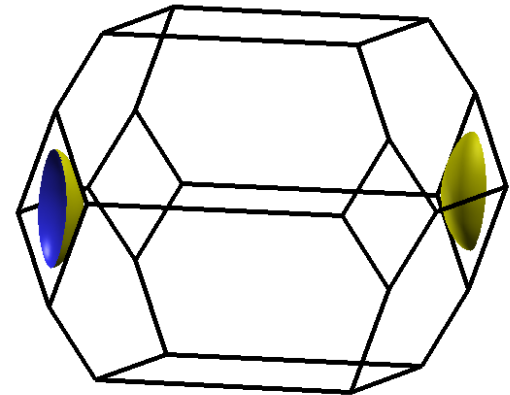
Energy Isosurfaces



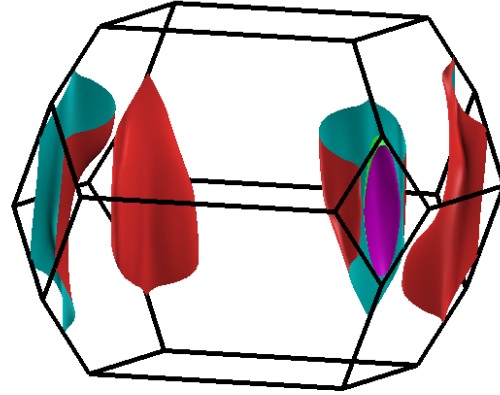
VBM - 0.05 eV



CBM + 0.05 eV



VBM - 0.1 eV

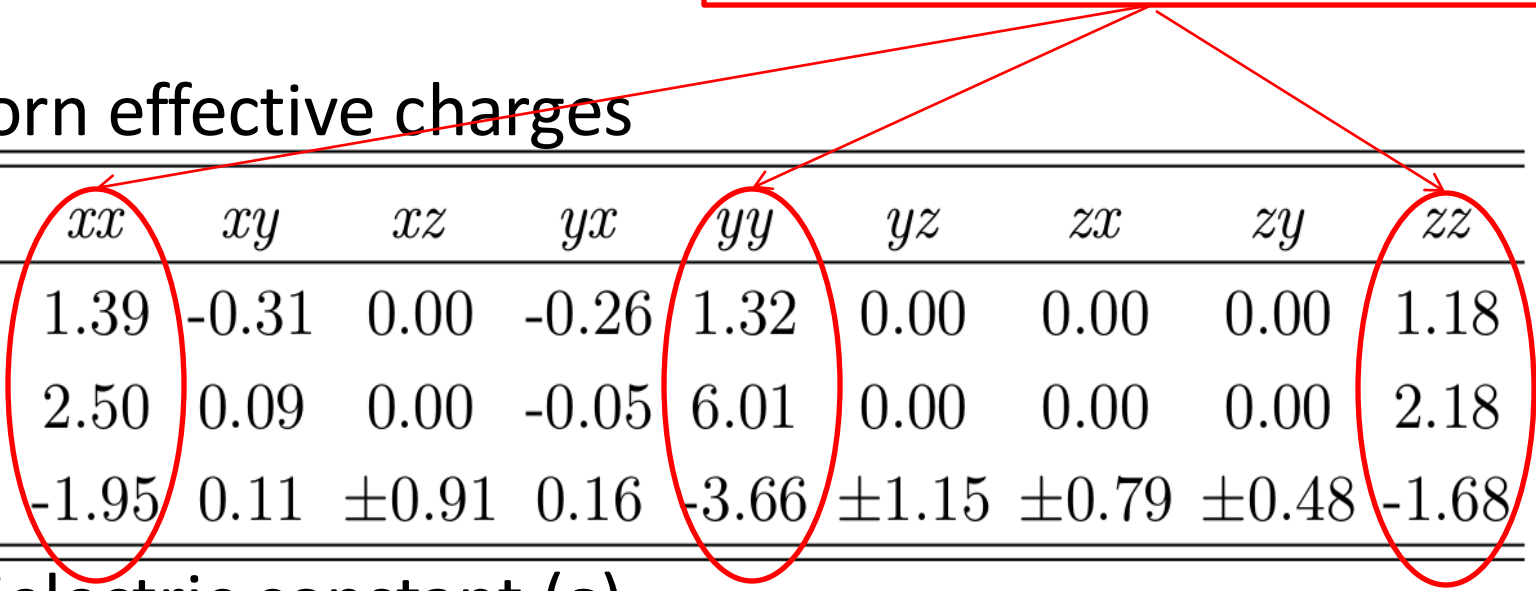


CBM + 0.1 eV

- Highly anisotropic carrier pockets
- Low dimensional, cylindrical around zone boundaries
- Direction dependent transport effective mass

Deviations from nominal valence

- Born effective charges



	xx	xy	xz	yx	yy	yz	zx	zy	zz
Na	1.39	-0.31	0.00	-0.26	1.32	0.00	0.00	0.00	1.18
Sb	2.50	0.09	0.00	-0.05	6.01	0.00	0.00	0.00	2.18
Se*	-1.95	0.11	± 0.91	0.16	-3.66	± 1.15	± 0.79	± 0.48	-1.68

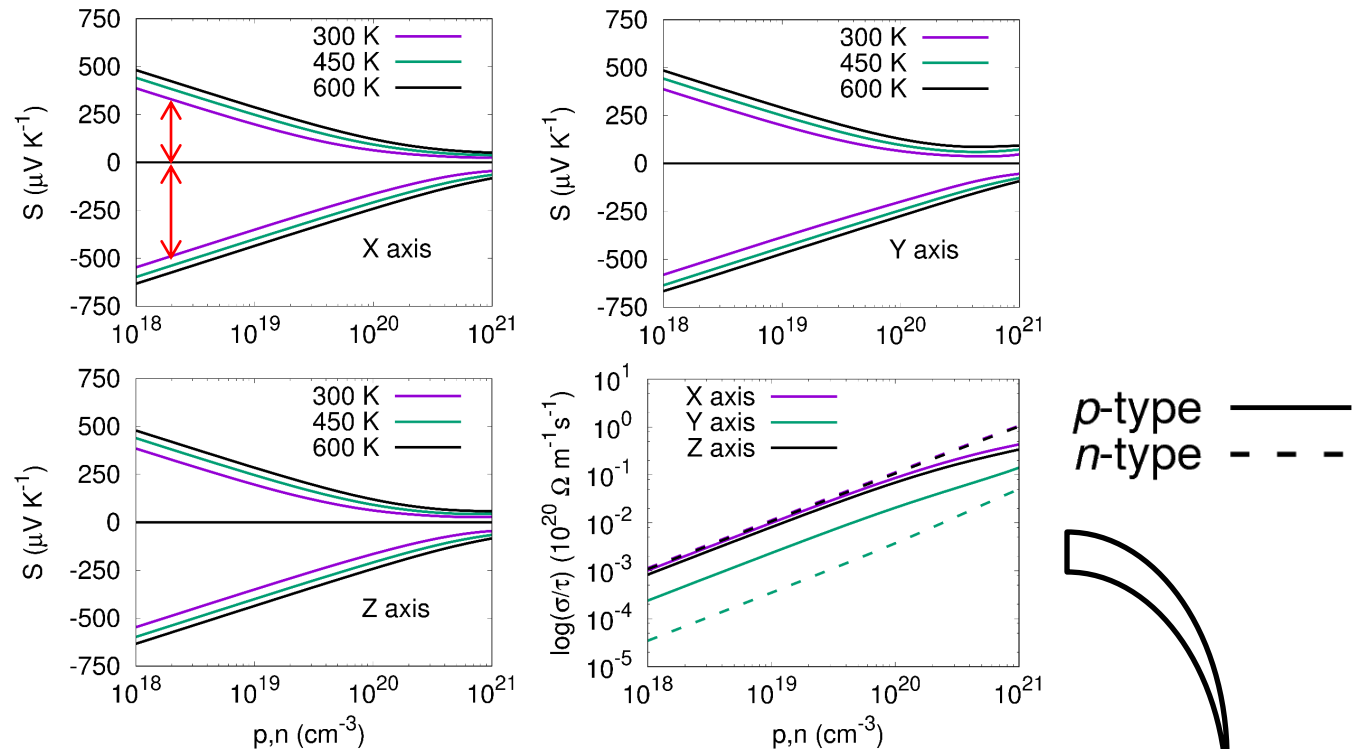
- Dielectric constant (ϵ)

	xx	xy	yy	zz
$\epsilon_{\infty,ij}$	9.2	1.4	13.1	9.6
$\epsilon_{ph,ij}$	9.4	-14.1	60.1	3.3
ϵ_{ij}	18.6	-12.7	73.2	12.9

- Highly anisotropic ionic part
- Very high ϵ in y direction

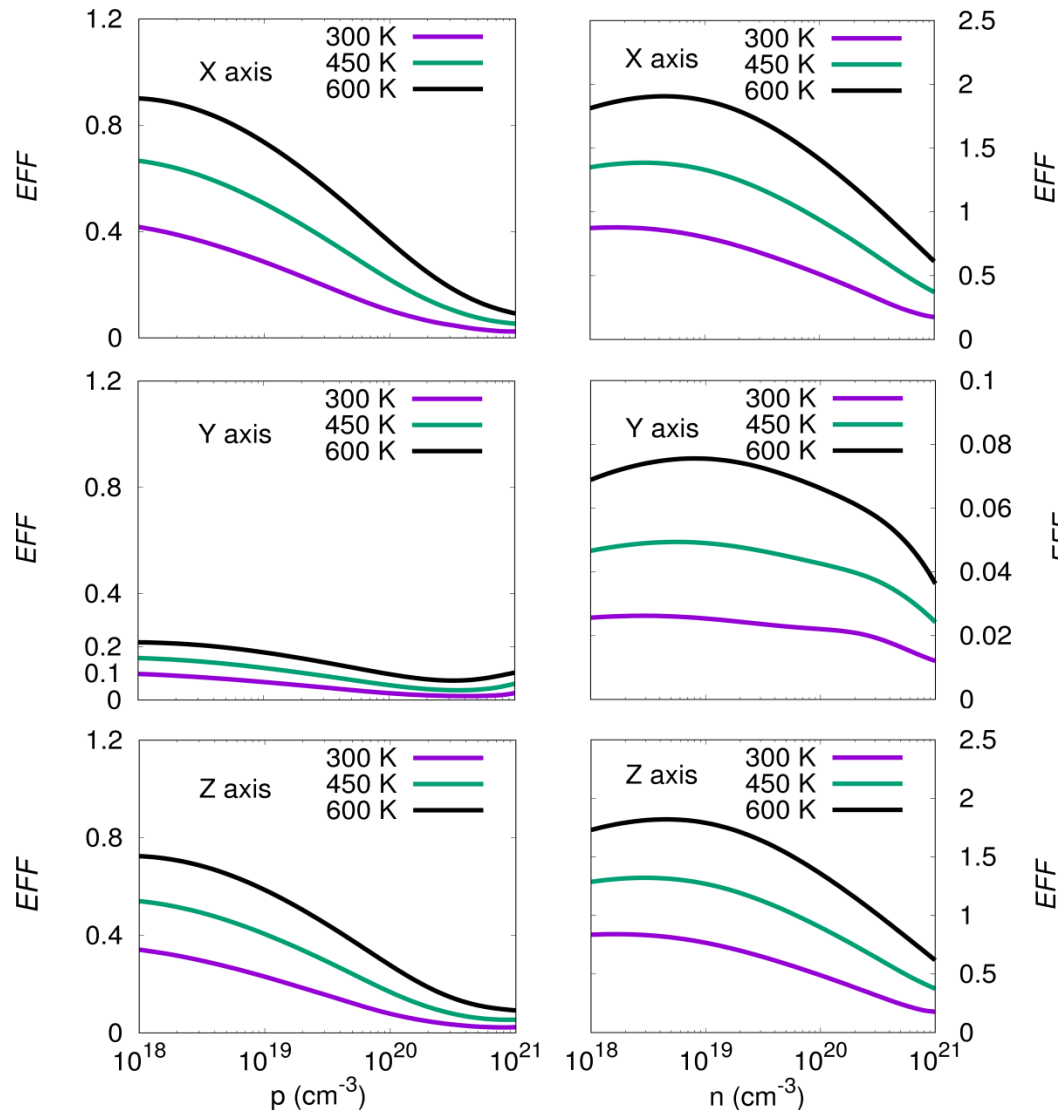
- Defect-tolerance $\rightarrow$ High mobility

Thermoelectric properties



- $S_{n\text{-type}} > S_{p\text{-type}}$ ($\sim 300 \mu\text{V K}^{-1}$)
- Larger S for interplanar y direction
- Even larger conductivity (reduced, σ/τ) anisotropy

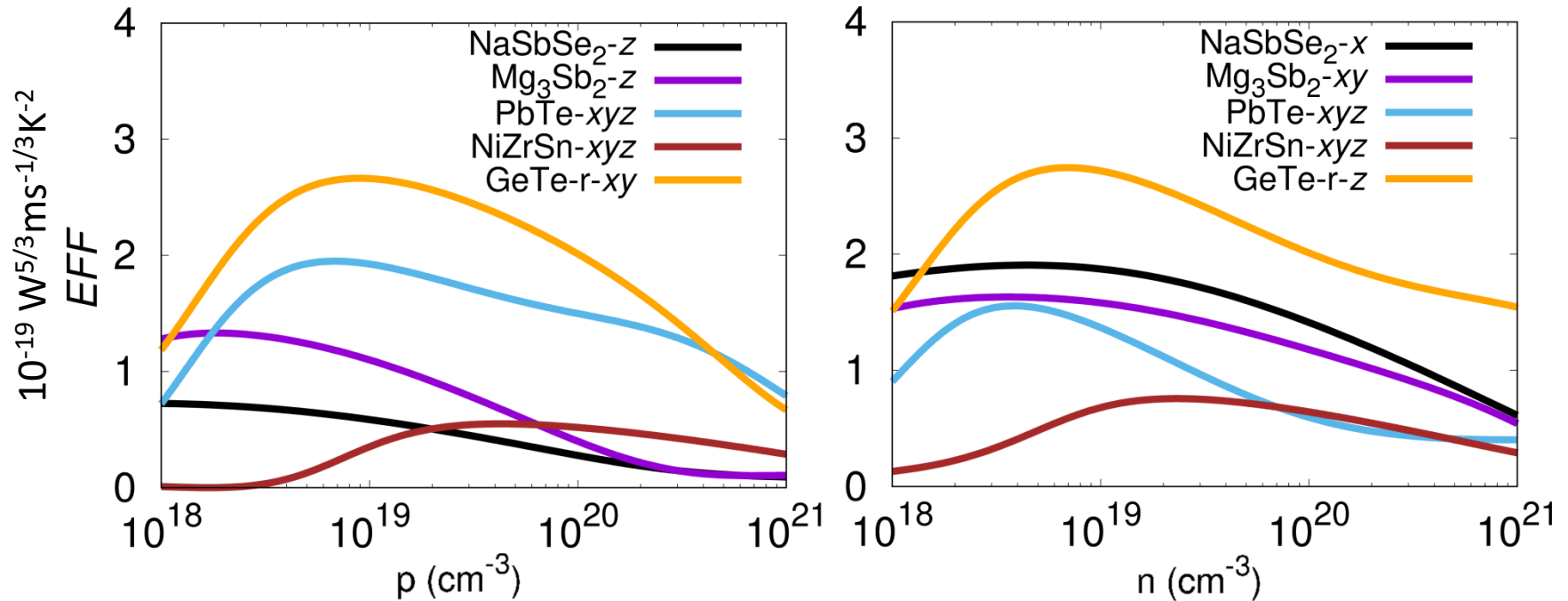
Electron Fitness Function (t)



- $t^\ddagger = (\sigma/\tau)S^2/N^{2/3}$
- t : Decoupling of S and σ
- Anisotropic transport properties
- In-plane/Out-of-plane variation in transport
- Favorable optimized carrier concentration

‡ Refer: *Xing et al*, Phys. Rev. Mater., **1**, 065405 (2017)

Benchmarking @ 600K



- *n*-type has superior performance
- Comparable to GeTe (electronic part) and better than known TE Mg₃Sb₂ ($ZT = 1.5$).

Summary

- Monoclinic layered structure with
 - Sufficient band gap → Negligible bipolar effect
 - High dielectric constant → High mobility
 - Large Seebeck + Good conductivity → Better transport
 - Complex, low dim. Fermi surface → favorable band structure
- Possibility of an efficient thermoelectric for n-doped NaSbSe_2

THANK YOU